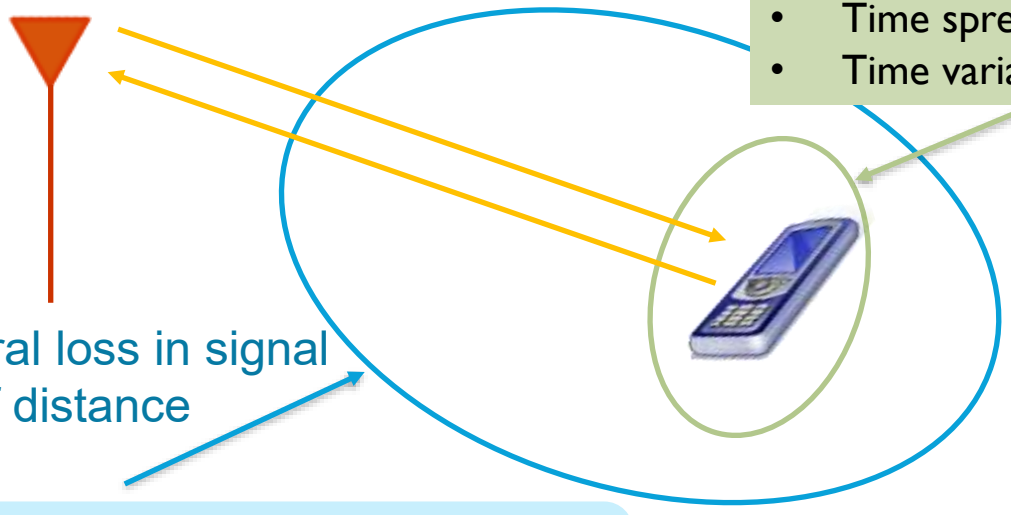
Lecture #16

EE 471C / EE 381K-17 Wireless Communication Lab

From last lecture

Captures more subtle multi-path effects

Small scale fading
- Time spreading of channel
- Time variance of channel



Captures general loss in signal as a function of distance

Large scale fading ≈ 100's λ
- Variation of mean signal strength versus distance
- Variation about mean

Preview of today's lecture

- Small scale fading overview
  - Frequency and time selective fading
  - Power delay profile and Doppler spectrum
- Frequency selective fading
  - Frequency selective channel versus flat cannel
  - Define power delay profile, RMS delay spread, coherence bandwidth
- Time selective fading
  - Time selective channel versus time invariant channel
  - Define Doppler power spectrum, Doppler spread, coherence time
- Small-scale channel models

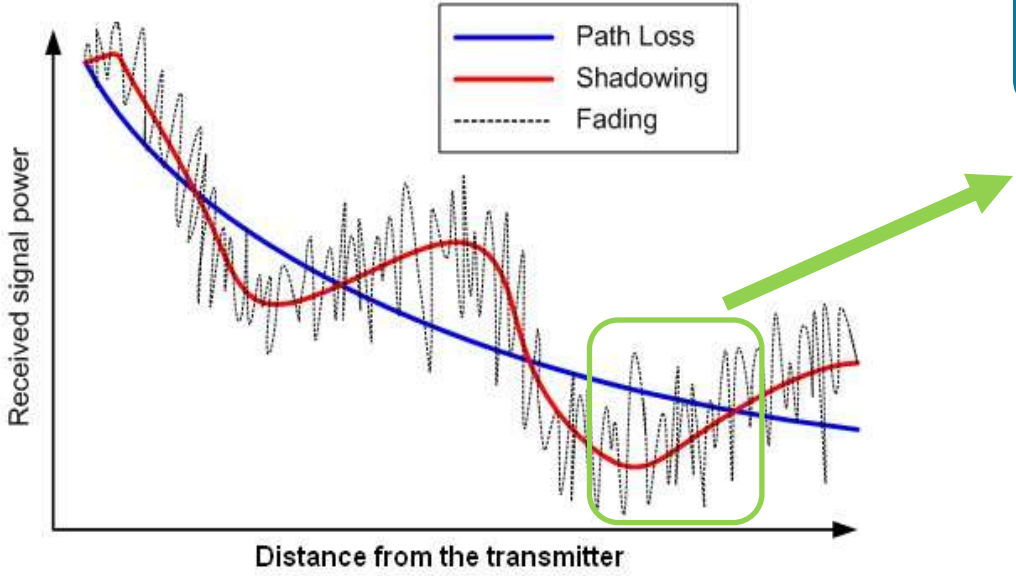
Small Scale Fading Overview

Learning objectives

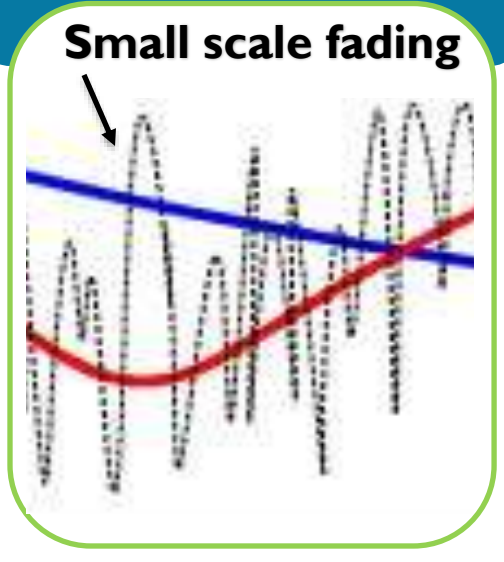
- Objective: Describe the factors that influence small-scale fading

Small scale fading

- Fluctuation in the envelope of the received signal over small distance (on the order of a wavelength)
- Due to the constructive and destructive interference of multipath components

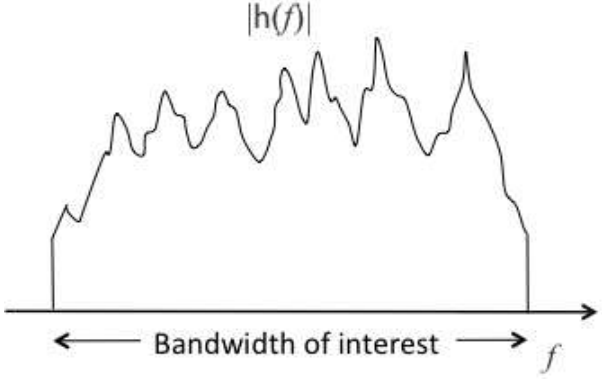


Effects occur on the order of wavelength distance

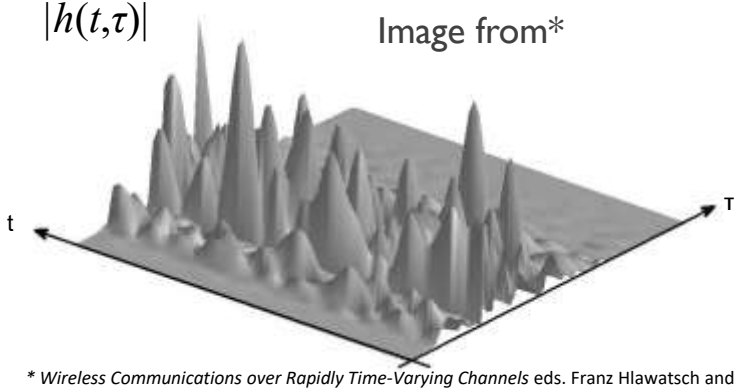


Frequency and time selective fading

- **Small scale fading** manifests in two forms of selectivity that are assumed to be independent of each other
- **Frequency-selective fading**: smearing of Rx signal due to multipath
  - As the multipath intensity increases, the channel will vary more sharply as a function of frequency



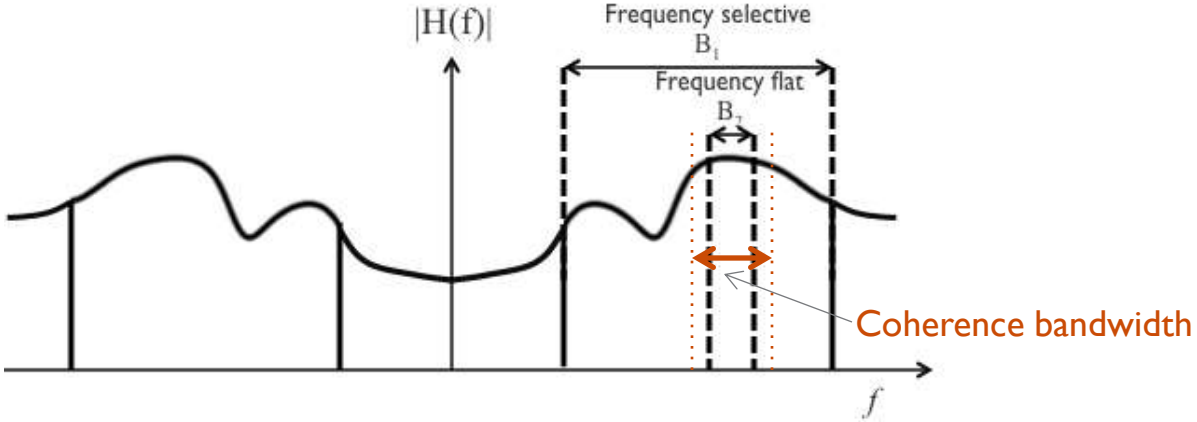
- **Time-selective fading**: variation of channel over time due to mobility of Tx and/or Rx



** Wireless Communications over Rapidly Time-Varying Channels* eds. Franz Hlawatsch and Gerald Matz, The Netherlands: Academic Press, Chapter 1, pp. 1–63

Frequency-selective fading

- Variation of the channel amplitude with respect to frequency
- **Coherence bandwidth**
  - Range of frequencies over which the channel amplitude remains fairly constant
  - Function of the extent and spread of multipath of the channel
- If the signal bandwidth is less than the coherence bandwidth the channel can be modeled as frequency flat
- Frequency selectivity is measured using RMS delay spread



Power delay profile

Complex BB equivalent channel $h(t,\tau)$ (time, lag)

- The wide sense stationary uncorrelated scattering assumption (WSSUS) leads to decoupled time and frequency selectivity
- Assume a WSSUS doubly selective complex BB equivalent channel

$h(t,\tau)$ — time-delay domain

Fourier transform of $h(t,\tau)$ with respect to the time domain

$H(\tau, f)$ — Doppler frequency domain

$$\mathbb{E}\left[H(\tau, f)H^*(\tau + \Delta_{\text{lag}}, f + \Delta_{\text{freq}})\right] = \delta(\Delta_{\text{lag}})\delta(\Delta_{\text{freq}})S_{\text{delay/Doppler}}(\tau, f)$$

delay-Doppler power spectrum

- Power delay profile

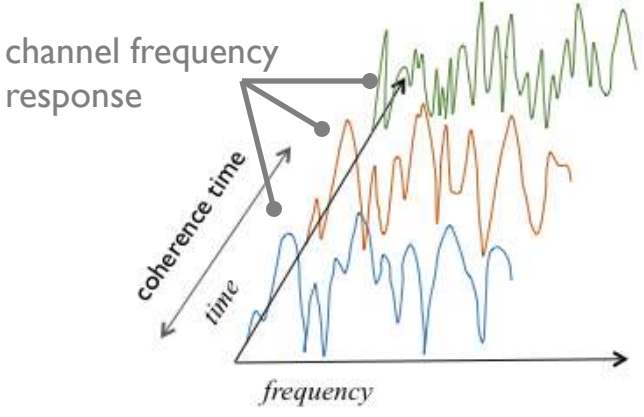
$$R_{\text{delay}}(\tau) = \mathsf{S}_{\text{delay/Doppler}}(\tau, 0)$$

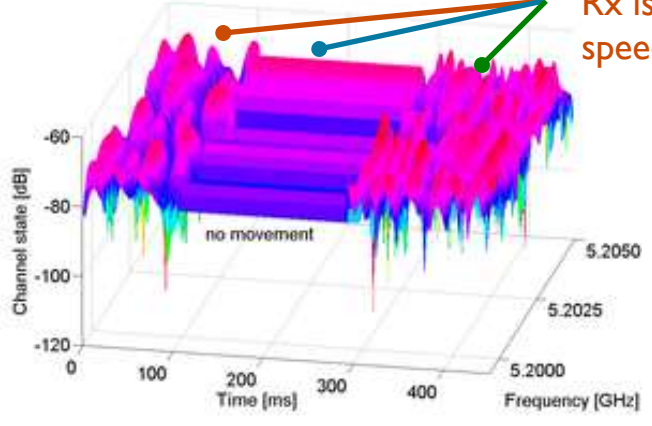
Measures how the energy of the channel varies in the lag domain for two very short signals

Decides on frequency selectivity

Time-selective fading: coherence time

- **Coherence time**: time duration over which the channel remains fairly constant
- The coherence time provides guidance on when the channel model can be assumed to be LTI
  - If the duration of the processing block in the RX is lower than the coherence time, the channel can be assumed LTI



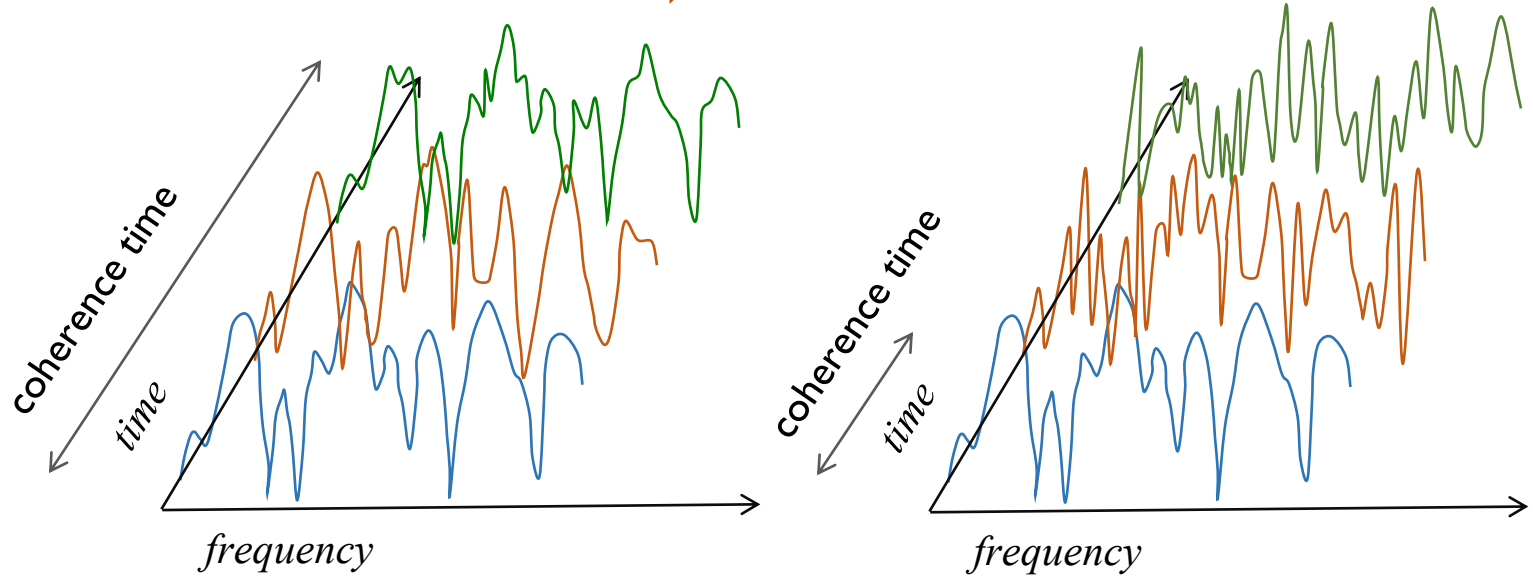


Example of a wireless channel generated with ChSim*

* S. Valentin, "ChSim -- A wireless channel simulator for OMNeT++", in TKN TU Berlin Simulation workshop, September 2006. Software and documentation available at http://wwwcs.upb.de/cs/chsim.

Time-selective fading: fast and slow fading

- Greater the time selectivity, the lower the coherence time of the channel
  - Low coherence times ⟹ **fast fading** or time variant channels
  - High coherence times ⟹ **slow fading** or time invariant channels



- Time selectivity is a function of the mobility present in the channel and is measured using the Doppler spread

Doppler spectrum

- For a WSSUS doubly selective complex BB equivalent channel

$h(t,\tau)$ — time-delay domain

Fourier transform of $h(t,\tau)$ with respect to the time domain →

$H(\tau, f)$ — Doppler frequency domain

delay-Doppler power spectrum

$$\mathbb{E}\left[H(\tau, f)H^*(\tau + \Delta_{\text{lag}}, f + \Delta_{\text{freq}})\right] = \delta(\Delta_{\text{lag}})\delta(\Delta_{\text{freq}})S_{\text{delay/Doppler}}(\tau, f)$$

- Doppler spectrum

Measures how much power is present in two different Doppler shifts of two narrowband signals

$$S_{\text{Doppler}}(f) = S_{\text{delay/Doppler}}(0, f)$$

Decides on time selectivity

Small Scale Fading Overview

Learning objectives

- Objective: Describe the factors that influence small-scale fading

Frequency Selective Fading

Learning objectives:

- Determine when a given propagation environment is frequency selective or frequency flat

Frequency selective channel versus flat channel

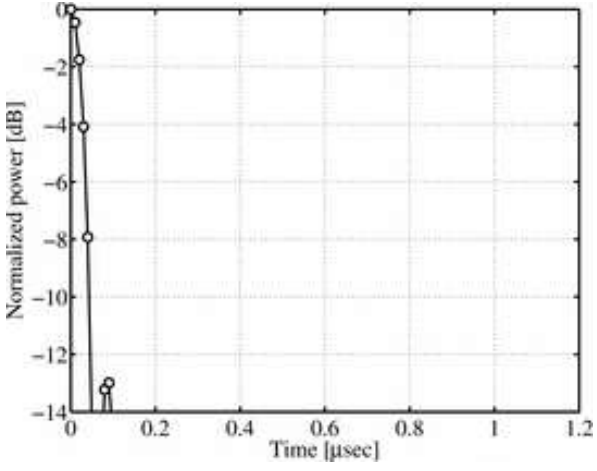
- Is a given channel frequency selective or frequency flat?
  - This depends on the signal itself
  - If the signal bandwidth is small, the symbol period is large and delays have to be larger to create frequency selectivity
- The most common approach in wireless is to base this decision on the **multipath intensity profile** or **power delay profile**

$$R_{\text{delay}}(\tau) = \mathsf{S}_{\text{delay/Doppler}}(\tau, 0)$$

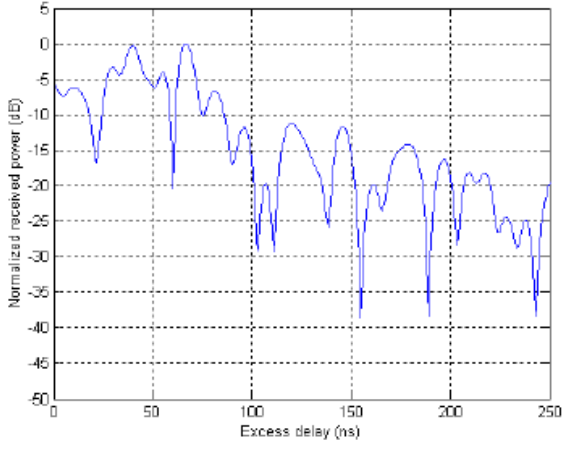
- The power delay profile is typically determined from measurements

Power delay profile

- A flat channel should have a profile that looks like a delta function
- A selective channel has more of a box-like profile



Flat channel*



Frequency selective channel**

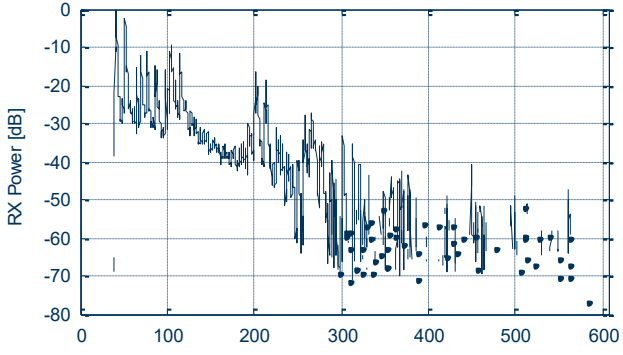
*H. Hatamoto et al, A Study on Power Delay Profile Measurement Using IEEE 802.11g Based Long Preamble Signals for WLAN Systems at 2.4 GHz Band", Vehicular Technology Conference (VTC Spring), 2012

**M. Carroll and T. A. Wysocki, "Delay Characteristics for an IEEE 802.11a Indoor Wireless Channel", Mobile Future and Symposium on Trends in Communications, 2003.

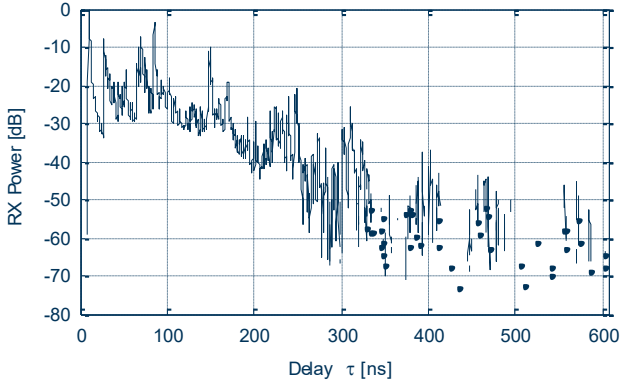
Measuring the power delay profile

- Example of channel sounding for the new WiFi standard IEEE 802.11ay*

Power Delay Profile Rx 1



Power Delay Profile Rx 3



* **"Channel Sounding for 802.11ay", doc: IEEE 802.11.15/0879r1,** 2015-07-14

Metrics

- Mean-excess delay $$\bar{\tau} = \frac{\int_0^\infty R_{\text{delay}}(\tau)\tau \mathrm{d}\tau}{\int_0^\infty R_{\text{delay}}(\tau)d\tau}$$
- Second moment $$\overline{\tau^2} = \frac{\int_0^\infty R_{\text{delay}}(\tau)\tau^2 d\tau}{\int_0^\infty R_{\text{delay}}(\tau)d\tau}$$
- Root mean square (RMS) delay spread $$\sigma_{\text{RMS,delay}} = \sqrt{\overline{\tau^2} - (\bar{\tau})^2}$$

Measures spikeyness or flatness of the power delay profile

- The coherence bandwidth is usually defined as $$B_{\text{coh}} = \frac{1}{5\sigma_{\text{RMS,delay}}}$$
- Metrics are often tabulated in measurement papers

Frequency selective channel versus flat channel

- A channel is said to be frequency flat if the symbol period satisfies

$$T \gg \sigma_{\text{RMS,delay}}$$

- Equivalently frequency flat if the bandwidth of the signal satisfies

$$B \ll B_c$$

- Otherwise the channel is frequency selective

Spaced-frequency correlation function

- Fourier transform of the power delay profile
- Measures the correlation as a function of the difference between subcarriers

$$\mathrm{S}_{\mathrm{delay}}(\Delta_{\mathrm{lag}}) = \int_0^{\infty} R_{\mathrm{delay}}(\tau) e^{-j2\pi\Delta_{\mathrm{lag}}\tau} \mathrm{d}\tau$$

- Measure by sending (short) sinusoids at f_2 and f_1, $\Delta_{lag} = f_2 - f_1$ and estimating the correlation between their respective channels
- One definition of coherence bandwidth is the smallest Δ_{lag} such that

$$|\mathrm{S}_{\mathrm{delay}}(B_{\mathrm{coh}})| = 0.5\mathrm{S}_{\mathrm{delay}}(0)$$

Frequency Selective Fading

Learning objectives:

- Determine when a given propagation environment is frequency selective or frequency flat

Time Selective Fading

Learning objectives:

- Determine when a given propagation environment is time selective or non-time selective

Time selective versus non-time selective channels

- Is is a given channel time selective?
  - This depends on the signal itself
  - If hundreds of symbols can be sent before the channel changes then probably it can be modeled as slowly varying for those symbols
- Often measured using the coherence time of the channel T_c
  - Time over which the channel remains fairly constant
- The coherence time is defined from the Doppler spectrum
- Stochastic models are also used to understand the temporal variability of the channel

Doppler power spectrum

- Measures energy as a function of frequency

$$S_{\text{Doppler}}(f) = S_{\text{delay/Doppler}}(0, f)$$

- *Doppler spread* can be defined in a similar way as RMS delay spread

$$\sigma_{\text{RMS,doppler}}$$

- In mobile channels we often use the maximum Doppler shift (as a worst case)

$$f_m = \frac{v f_c}{c}$$

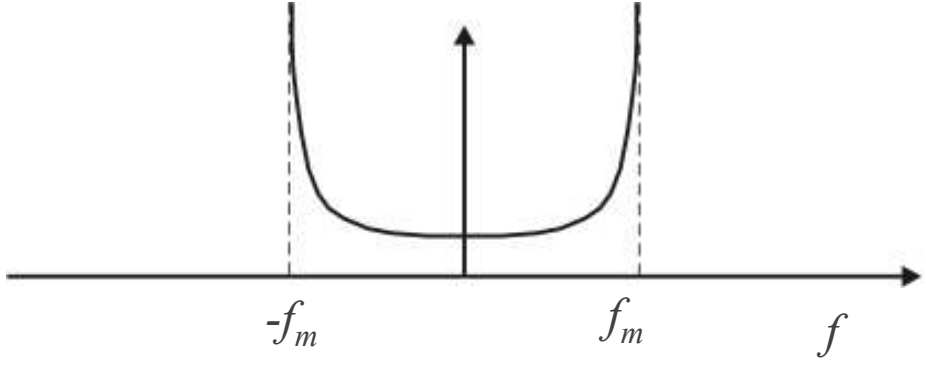
time selective fading depends on the carrier

- Maximum Doppler shift gives a reasonable approximation of RMS Doppler spread

Clarke-Jakes model for the Doppler spectrum

- The most common **analytical model** for the Doppler spectrum
- The TX is stationary while the RX is moving with a velocity v towards the TX

$$S_{\text{Doppler}}(f) = \begin{cases} \frac{1}{\pi\sqrt{f_{\text{m}}^2 - f^2}} & \nu \in [-f_{\text{m}}, f_{\text{m}}] \\ 0 & \nu \notin [-f_{\text{m}}, f_{\text{m}}] \end{cases}$$



Spaced-time correlation function

- It is the inverse Fourier transform of Doppler spectrum

$$R_{\text{Doppler}}(\Delta_{\text{time}}) = \int_{-\infty}^{\infty} \mathsf{S}_{\text{Doppler}}(f) e^{j2\pi\Delta_{\text{time}} f} \mathrm{d}f.$$

- The space-time correlation function gives the correlation between narrowband signals at two different points in time Δ_{time}
- The coherence time of the channel is measured as the smallest value of Δ_{time} such that

$$|R_{\text{Doppler}}(\Delta_{\text{time}})| = 0.5 R_{\text{Doppler}}(0)$$

- The resulting value is T_{coh}

Time selective versus time invariant channels

- A channel can be assumed to be time invariant (not time selective) over block Ntot if

$$TN_{\text{tot}} \ll T_{\text{coh}}$$

- For the Clarke-Jake´s spectrum is common to take

$$T_{\text{coh}} = 0.423/f_{\text{m}}$$

- Estimate coherence time using either the maximum expected Doppler frequency or by estimating the coherence time directly

Time Selective Fading

Learning objectives:

- Determine when a given propagation environment is time selective or non-time selective

Signal models for channel selectivity

Learning objectives:

- Determine channel models for the different selectivity regions

Time invariant models

- Time invariant/frequecy flat

$$y[n] = e^{\mathrm{j}2\pi\epsilon n} hs[n-d] + v[n]$$

Equivalent system including frame delay and frequency offset

- Time invariant/frequency selective

Includes effects of multipath, TX pulse shape and RX matched filter

$$y[n] = e^{\mathrm{j}2\pi\epsilon n} \sum_{\ell=0}^{L} h[\ell] s[n-d-\ell] + v[n]$$

Equivalent system including the channel, frame delay and frequency offset

Time variant models

- Time variant/frequecy flat

$$y[n] = h[n]s[n-d] + v[n]$$

$$n = 0, 1, \ldots, N_{\text{tot}} - 1$$

The channel is changing slowly with respect to the symbol period but is changing faster than $T N_{\text{tot}}$

- Time variant/frequency selective

Two-dimensional time varying system

$$y[n] = \sum_{\ell=0}^{L} h[n, \ell]s[n-d-\ell] + v[n]$$

$$n = 0, 1, \ldots, N_{\text{tot}} - 1$$

Most challenging from a SP perspective

Signal models for channel selectivity

Learning objectives:

- Determine channel models for the different selectivity regions

Small scale channel models

Learning objectives:

- Describe the most common models for small scale fading

Frequency flat channel models: Rayleigh model

- h_{s} has distribution $\mathcal{N}_{\mathbb{C}}(0,1)$
- The variance is selected so that

$$h[\ell] = \sqrt{G} h_{\mathrm{s}}[\ell] \quad \text{for} \quad \ell = 0, 1, \ldots, L$$

small-scale

$$\mathbb{E}[|h_{\mathrm{s}}|^2] = 1$$

- Magnitude of the channel has Rayleigh distribution

$$f_{h_{\mathrm{s}}}(x) = 2x \exp(-x^2)$$

- It is used when there is a rich scattering NLOS environment
- Paths arrive from different directions with slightly different phase shifts

Frequency flat channel models: Ricean model

- If there is a dominant LOS path besides the rich NLOS scattering
- h_s has distribution $\mathcal{N}_{\mathbb{C}}(\mu, \sigma^2)$
- The Rice factor is

$$K = |\mu|^2/\sigma^2$$

- In terms of K h_s has distribution

$$\mathcal{N}_{\mathbb{C}}(e^{\theta}\sqrt{K/(1+K)}, 1/(1+K))$$

$$\theta = \text{phase}(\mu)$$

Frequency selective channel models: Rayleigh

- $h_s[l]$ is distributed as $\mathcal{N}_{\mathbb{C}}(0, R_{\text{delay}}[\ell])$
  - Special case: uniform power delay profile

    $h_s[l]$ is distributed as $\mathcal{N}_{\mathbb{C}}(0, 1)$

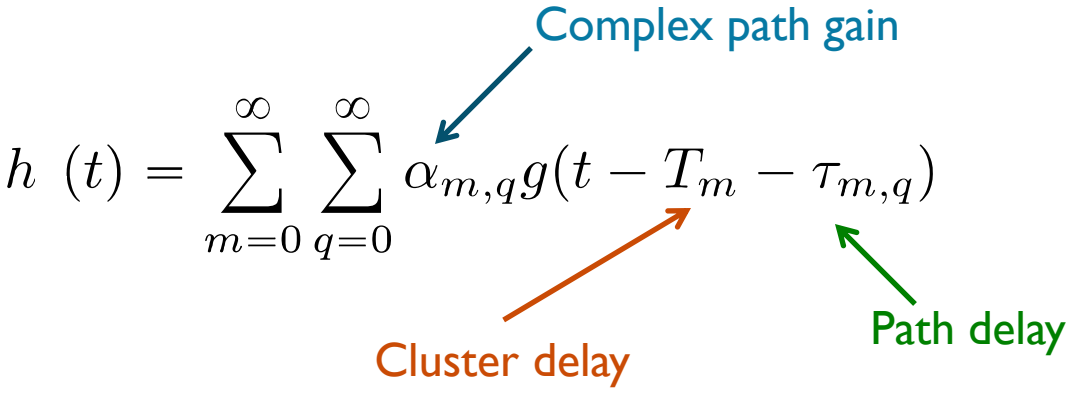
- The power in the channel is

$$\sum_{\ell=0}^{L} |h_s[\ell]|^2 = \sum_{\ell=0}^{L} R_{\text{delay}}[\ell]$$

  - May be greater than one
  - Realistic because the multipath allows the RX to capture more energy

Saleh-Valenzuela model

- Complex BB equivalent model



- Inspired by measurements that show that multipath tend to arrive in clusters
- Statistical models for the different parameters

Performance analysis with flat fading channels-I

$$y[n] = \sqrt{E_{\mathrm{x}}} h_{\mathrm{s}} s[n] + v[n]$$

- The instantaneous probablity of symbol error is a RV

$$P_e\left(\frac{E_{\mathrm{x}}}{N_{\mathrm{o}}}|h[n]\right)$$

- Performance measure is the average probability of symbol error

$$P_{\mathrm{e}}\left(\frac{E_{\mathrm{x}}}{N_{\mathrm{o}}}\right) = \mathbb{E}_{h_{\mathrm{s}}}\left[P_{\mathrm{e}}\left(\frac{E_{\mathrm{x}}}{N_{\mathrm{o}}}|h_{\mathrm{s}}\right)\right] \qquad P_{\mathrm{e}}\left(\frac{E_{\mathrm{x}}}{N_{\mathrm{o}}}\right) = \int_c P_{\mathrm{e}}\left(\frac{E_{\mathrm{x}}}{N_{\mathrm{o}}}|c\right) f_{h_{\mathrm{s}}}(c)\mathrm{d}c$$

- Using the union bound

Using the Chernoff bound $Q(x) \leqslant e^{-\frac{x^2}{2}}$

$$P_{\mathrm{e}}\left(\frac{E_{\mathrm{x}}}{N_{\mathrm{o}}}|h\right) \leqslant (M-1)Q\left(\sqrt{\frac{E_{\mathrm{x}}|h|^2}{N_{\mathrm{o}}}\frac{d_{\min}^2}{2}}\right) \leqslant \frac{1}{2}(M-1)e^{-\frac{E_{\mathrm{x}}|h|^2}{4N_{\mathrm{o}}}d_{\min}^2}$$

Performance analysis with flat fading channels-2

- Assuming that the channel has the Rayleigy distribution

$$\int_c \frac{1}{2}(M-1)e^{-\frac{E_x|c|^2}{4N_o}d_{\min}^2}\frac{1}{\pi}e^{-|c|^2}\mathrm{d}c$$

$$= \frac{1}{2}(M-1)\int_c e^{-\frac{E_x|c|^2 d_{\min}^2}{N_o}+|c|^2}\mathrm{d}c$$

$$= \frac{1}{2}(M-1)\int_c e^{-|c|^2\left(\frac{E_x|c|^2 d_{\min}^2}{N_o}+1\right)}\mathrm{d}c$$

$$= \frac{1}{2}(M-1)\int_c \frac{\frac{E_x|c|^2 d_{\min}^2}{N_o}+1}{\frac{E_x|c|^2 d_{\min}^2}{N_o}+1}\mathrm{d}c$$

$$= \frac{1}{2}(M-1)\frac{1}{\frac{E_x|c|^2 d_{\min}^2}{N_o}+1}$$

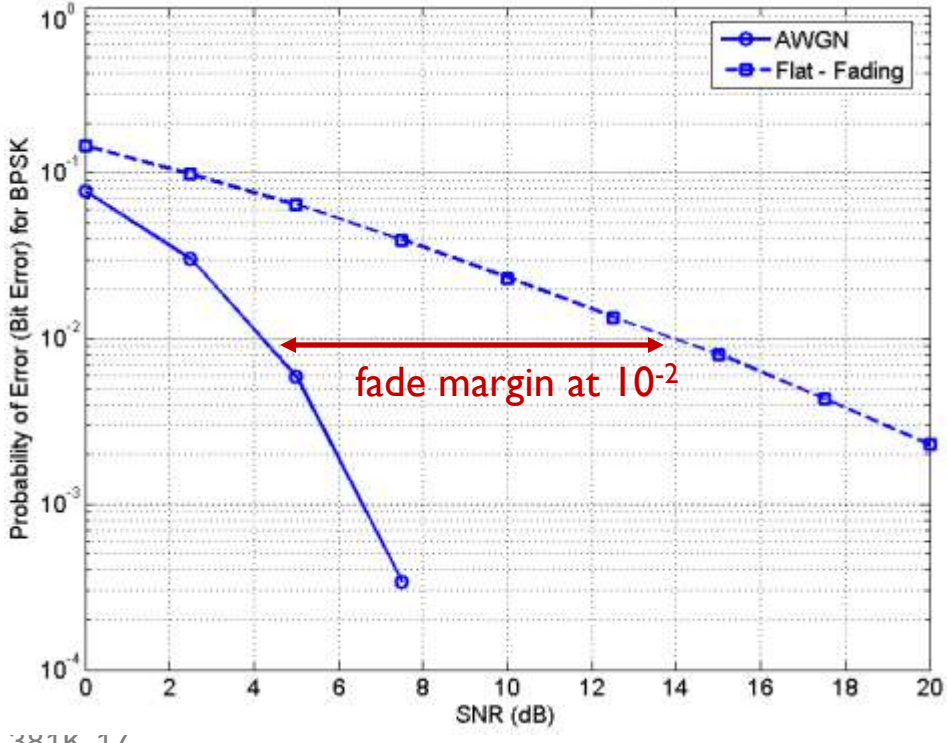
- Then

$$P_e\left(\frac{E_x}{N_o}|h\right) \leqslant \frac{1}{2}(M-1)\frac{1}{\frac{E_x|c|^2 d_{\min}^2}{N_o}+1}$$

the probability of error decreases as a function of the inverse of the SNR for Rayleigh fading channels

Performance analysis with flat fading channels-3

- Probability of error curve comparing Gaussian channel and Rayleigh fading for BPSK



Fading channels require a much higher average SNR to achieve a given probability of error

Small scale channel models

Learning objectives:

- Describe the most common models for small scale fading